UNDERGRADUATE PROJECT

AidField

Offline-First Emergency First Aid & AI Triage App

React Native · SQLite · Google Gemini API · Android & iOS

Kenya-focused · Offline-first · Voice & Image Input · English & Swahili

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Chapter 1 — Introduction

Background, problem, objectives & scope

Background & Problem Statement

1.1 Background

- Kenya rural emergency response times: 30–90 minutes — far beyond the golden hour
- 50%+ smartphone penetration, yet nearly all first aid apps require internet
- Standard guides assume a medical kit is present — no improvised resource guidance
- AidField: offline-first native app (React Native) + AI triage (Gemini API)

1.2 Problem Statement

No connectivity, no help

Existing apps fail offline — precisely when emergencies occur far from help.

No improvised guidance

No app teaches belts, sticks, or cloth as substitutes for medical supplies.

Panic & paralysis

Untrained bystanders freeze. No calm, voice-guided, step-by-step interface exists.

Aim & Research Objectives

Aim

Design and develop AidField — an offline-first, AI-assisted downloadable mobile application providing emergency first aid guidance with improvised resource recommendations, multimodal input, and graceful offline degradation.

Specific Objectives

1. Investigate

First aid knowledge gaps and emergency response challenges in low-connectivity Kenyan environments.

2. Design

A panic-proof, user-centred interface minimising cognitive load, with text, voice, and camera input.

3. Develop

AI triage module using Gemini API for plain-language, speech-transcribed, and image-based emergency descriptions.

4. Implement

Improvised resource engine mapping household materials to standard first aid supply substitutes.

5. Evaluate

Usability, accuracy, and effectiveness through user testing with representative target users.

Significance, Scope & Key Limitations

1.4 Significance

Social & Health

40% of trauma deaths are preventable with timely first aid (WHO). AidField bridges the gap before help arrives in rural Kenya.

Technological

Offline-first AI in native mobile; advances mHealth design using React Native, SQLite, Gemini API, and native speech/camera APIs.

Academic

Contributes to HCI in crisis scenarios; aligns with SDG 3 (Good Health & Well-Being); replicable framework for health apps in developing countries.

1.5 Scope

- Android (APK sideload) & iOS (TestFlight) — no app store account required for Android
- 50 offline scenarios: bleeding, burns, choking, fractures, snake bites, cardiac arrest & more
- English & Swahili · Kenya emergency contacts pre-loaded (999, 112, Red Cross)
- Geographically focused: Nairobi, Kiambu, Mombasa, Nakuru counties

1.7 Key Limitations

- AI accuracy for rare presentations → 90% threshold enforced via prompt engineering
- Offline AI (voice/image) requires connectivity → 50 cached scenarios always available
- Risk of delaying professional care → persistent one-tap emergency dial on every screen
- iOS without App Store → TestFlight during development; App Store planned post-project

Chapter 2 — Literature Review

Literature Review

Why existing solutions fall short

Millions die each year from emergencies where immediate action could have helped. Smartphones reach 6.8 billion people — yet most first aid apps remain static, internet-dependent, and hard to use under stress. AidField was designed to close this gap.

4.4M

annual preventable injury deaths (WHO)

6.8B

smartphone users worldwide

0

apps with AI-driven dynamic guidance

Related System 1 — American Red Cross First Aid App

Strengths

- Step-by-step guidance for burns, bleeding, choking, cardiac arrest
- Partial offline access for a subset of content
- Over 1 million downloads, medically endorsed
- Includes videos, quizzes, and hospital locator

Limitations

- Static content only — cannot handle unusual scenarios
- No voice input or image-based injury assessment
- Calibrated for North American users and contexts
- Offline coverage is partial, requiring re-synchronisation

Related System 2 — St John Ambulance First Aid App

Strengths

- Clinically reviewed, evidence-based guidance
- Visual aids — photos and diagrams per step
- CPR metronome for compression timing
- More comprehensive offline coverage than most competitors

Limitations

- All content is pre-authored — no dynamic or AI guidance
- No voice or camera input for hands-free use
- No improvised resource suggestions for low-resource settings
- Cannot interpret ambiguous or multi-symptom scenarios

Related System 3 — ICE (In Case of Emergency) Apps

Strengths

- Stores blood type, allergies, medications, and contacts
- Accessible to first responders even when device is locked
- Improves outcomes for patients with complex medical histories
- Simple, low-maintenance setup for end users

Limitations

- Passive — displays information but gives no procedural guidance
- Assumes a trained responder is present to act on data
- Offers minimal value to untrained bystanders
- Does not respond to or interpret the current emergency

How AidField Addresses Key Weaknesses

Static content

Gemini API generates real-time, tailored guidance for any scenario described in natural language.

No voice / image input

Voice-in via speech recognition; camera images sent to Claude's vision API for injury assessment.

Incomplete offline

50 scenarios cached on install. Full guidance available with zero connectivity, always.

Geographic narrowness

AI adapts to local context, resources, and hazards when the user provides location detail.

Chapter 3 — Methodology

How AidField will be built

Selected Methodology — Agile Iterative Development

AidField's technical complexity — AI prompt refinement, dual-tier architecture, and high usability demands — makes an iterative approach the right fit. Each sprint produces a testable increment.

AI Integration

Gemini API outputs require iterative prompt tuning — waterfall would not allow this.

Usability Under Stress

High-stakes UX must be tested with real users and refined each cycle.

Modular Architecture

Offline and online tiers built and tested independently before integration.

Industry Alignment

Agile is standard in mobile app development for both Android and iOS.

Development Iterations

1. **Core App & Offline Library**

   App shell, navigation, 50 cached emergency scenarios with improvised resource tips. Fully functional offline.

2. **AI Integration**

   Google Gemini API connected. Online natural language query interface with graceful offline fallback.

3. **Voice & Image Input**

   Speech recognition for hands-free queries. Camera input for AI-assisted visual injury assessment.

4. **Testing & Refinement**

   Usability testing with 15 participants. Performance optimisation and app store submission preparation.

Data Collection

Primary

- Structured interviews with 10 participants — community health workers, first aid trainers, and laypersons
- Validates the 50 scenario selection and UI design priorities
- Usability testing with 15 participants using think-aloud protocol
- Metrics: task completion rate, time-on-task, error rate

Secondary

- Peer-reviewed literature in emergency medicine, medical informatics, and HCI
- Google API, Google Play, and Apple Developer documentation
- First aid protocols: Kenya Red Cross, WHO, and ILCOR guidelines

Tools & Technologies

React Native

Cross-platform — one codebase for Android & iOS

SQLite

Local offline storage for the 50-scenario library

Google Gemini API

Online AI guidance and image analysis (gemini-2.5-flash primary, gemini-1.5-flash fallback)

RN Voice / Camera

Native speech recognition and camera access

Jest + Detox

Unit and end-to-end automated UI testing

Git / GitHub

Version control with branch-based development

Chapter 4 — System Analysis

Understanding the problem & defining requirements

The Current System — How Bystanders Seek Help Today

Without AidField, a bystander in an emergency follows a fragmented, connectivity-dependent path:

Emergency Event

Injury, collapse, or sudden illness occurs

→

Seeks Help Online

Google Search, WebMD, NHS — all require internet

→

Or Calls Someone

Phone a friend, relative, or tries from memory

→

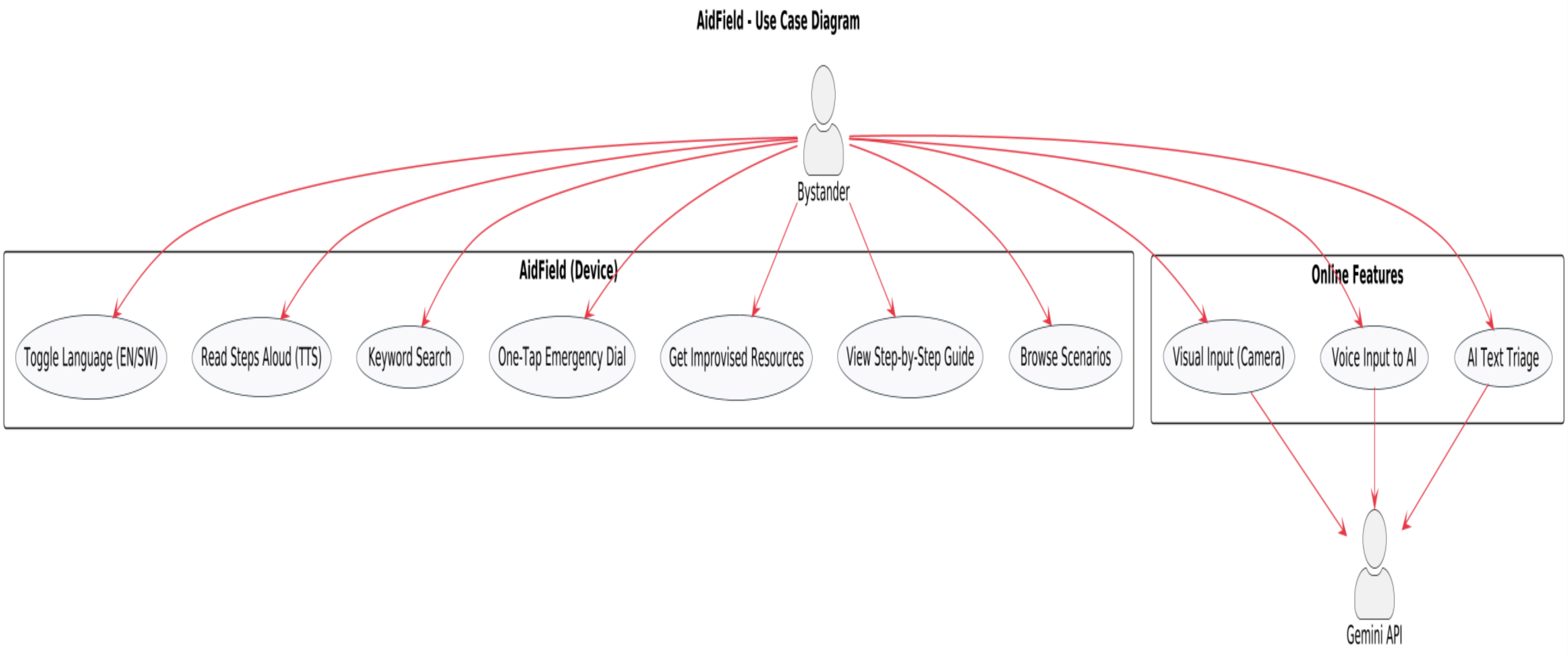
Outcome

Delayed, inconsistent, or no guidance at point of need

Core failures of the current system:

- All online options fail without connectivity
- No AI to interpret ambiguous or multi-symptom descriptions
- No improvised resource suggestions for low-resource settings

Fig 4.3 — Use Case Diagram: AidField System



Functional Requirements — Must Haves (MoSCoW)

15 functional requirements derived from Sprint 1 needs assessment (n=15). Key Must Have requirements:

FR01–02 · Offline Library

Cache 50 scenario guides on first launch, each with step-by-step instructions and improvised resource substitutes.

FR03–04 · Offline Routing

Route user text to closest scenario via keyword matching + decision tree. Show offline notice when AI unavailable.

FR05–06 · AI Triage

Submit description to Gemini API when online. Response must include action steps, urgency indicator, and improvised resources.

FR07–09 · Voice I/O

Capture voice input, transcribe via native speech API, and read instructions aloud (TTS) for fully hands-free operation.

FR10–11 · Image Input

Allow camera image for AI visual injury assessment when online. Hide and redirect offline.

FR12 · Emergency Calls

One-tap dialling of Kenya emergency numbers: 999, 112, Kenya Red Cross 0800 723 999.

Non-Functional Requirements — Key Quality Targets

Performance

Offline scenario loads ≤ 1 sec · AI response ≤ 5 sec on 4G

Usability

SUS score ≥ 68 · All critical actions within 3 taps from home

Reliability

Zero offline failures across 50 test runs · Graceful degradation within 2 sec of disconnect

Accuracy

AI triage ≥ 90% Correct or Partially Correct across 50 standardised test descriptions

Security

No PII transmitted — only anonymised emergency descriptions sent to API

Portability & Storage

Android 8.0+ and iOS 14+ · APK ≤ 50 MB · Installed ≤ 100 MB

Core Algorithms — How the System Routes Every Query

Offline Scenario Router

- Normalise query → lowercase, remove stopwords, stem tokens
- Match keywords against each scenario's keyword index
- If score ≥ 2 matched keywords → return top scenario
- If score < 2 → activate 6-node decision tree (conscious? bleeding? breathing?)
- Achieved 87% correct routing in Sprint 1 testing (n=30 queries)

Two-Tier Connectivity Controller

- NetInfo API polled every 2 seconds and on every user interaction
- Online → send text/speech to Claude text API, or image+text to Claude vision API
- Offline → hide AI features, show notice, route to offline scenario matcher
- Error states (429, 503, timeout) → silently fall back to offline guidance

Offline Dataset — 50 Emergency Scenarios

Validated against WHO (2020), Kenya Red Cross (2019), and ILCOR (2021) protocols.

Cardiovascular:	Cardiac arrest (adult & child), chest pain, stroke
Airway & Breathing:	Choking (adult & infant), drowning, asthma
Bleeding & Wounds:	Severe/minor bleeding, nosebleed, puncture, amputation
Burns & Scalds:	Superficial, partial, full-thickness, chemical, electrical
Fractures & Sprains:	Arm, leg, spinal, ankle, dislocation
Poisoning & Envenomation:	Snake bite, scorpion, bee sting, ingested/inhaled poison
Head & Neurological:	Head injury, seizures, febrile convulsion, concussion
Allergic Reactions:	Mild reaction, anaphylaxis
Environmental:	Heatstroke, hypothermia, dehydration, eye injuries
Obstetric & Paediatric:	Childbirth emergency, newborn resuscitation, infant fall
Trauma & Other:	RTA multiple injuries, crush injury, diabetic emergency

Chapter 5 — System Design

Architecture, database, UI & API integration

System Architecture — Offline-First, AI-Enhanced

All core guidance lives on-device. The Gemini API is an optional enhancement — activated only when online.

Presentation Layer

React Native UI — Home, Search, Scenario, Voice, Image, Settings screens

Connectivity Layer

NetInfo API — detects online/offline in real time, routes to correct tier

Offline Tier (always available)

SQLite 50-scenario DB · Keyword Index · Decision Tree · Improvised Resources · Swahili Store

Online Tier (when connected)

Google Gemini API — Text triage · Speech interpretation · Vision / image analysis

Device Hardware Layer

Microphone · Camera · Speaker · Local Storage

↕ *HTTPS / API calls when online*

Fig 5.1 — High-Level System Architecture

All core guidance lives on-device. Gemini API is an optional enhancement — active only when online.

[DELETE THIS TEXT - INSERT PLANTUML PNG DIAGRAM HERE] (PlantUML code is in the slide Speaker Notes) (React Native / Expo)

PRESENTATION LAYER

- Home Screen
- Search / AI
- Scenario
- Voice Input
- Image Input
- Settings

CONNECTIVITY LAYER

NetInfo API

Polls every 2 seconds

Routes to correct tier

OFFLINE TIER (Always On)

- SQLite DB (50)
- Keyword Index
- Decision Tree
- Improvised Res.
- Swahili Store

ONLINE TIER (When Connected)

- Gemini API
- Text triage
- Speech interp.
- Vision / image
- Streaming TTS

DEVICE HARDWARE   Microphone · Camera · Speaker · Local Storage

Google Cloud API (HTTPS — online only)

Database Design — 5-Table SQLite Schema

All offline data stored locally in SQLite. No cloud database required.

categories

category_id · category_name · icon_name

scenarios

scenario_id · FK category_id · title · keywords · urgency_level

steps

step_id · FK scenario_id · step_number · instruction · instruction_sw (Swahili)

imp_resources

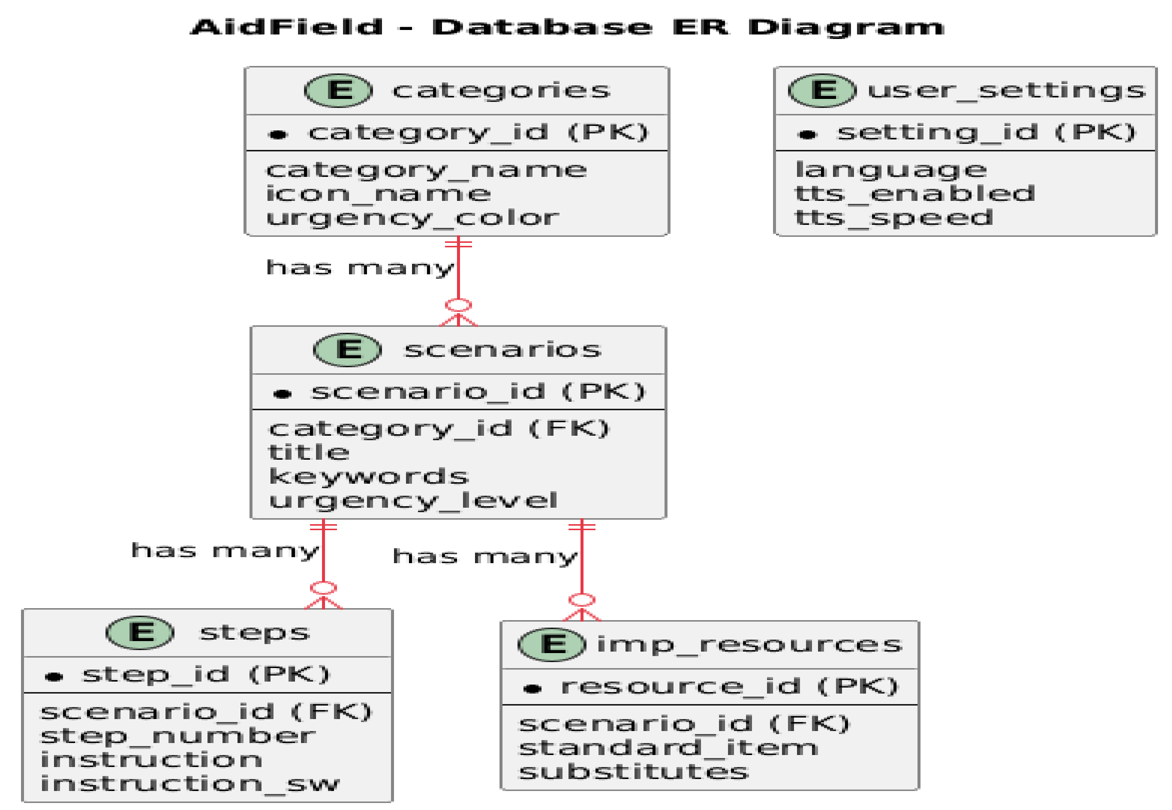
resource_id · FK scenario_id · standard_item · substitutes[ ] · notes

user_settings

setting_id · language (en/sw) · tts_enabled · last_scenario

categories 1—< scenarios 1—< steps & scenarios 1—< imp_resources

Fig 5.3 — Entity Relationship Diagram (ERD)



UI Design — Navigation & Key Screens

Stack-and-tab navigation. Designed for one-handed, high-stress use — every critical action within 3 taps.

Home

2×3 grid of colour-coded category buttons. Red = life-threatening, Amber = urgent, Teal = moderate. Floating emergency dial button always visible.

Search / AI

Free-text input + mic + camera icons. Offline: top-5 keyword matches refresh as user types. Online: Claude response card with steps, urgency badge, TTS button.

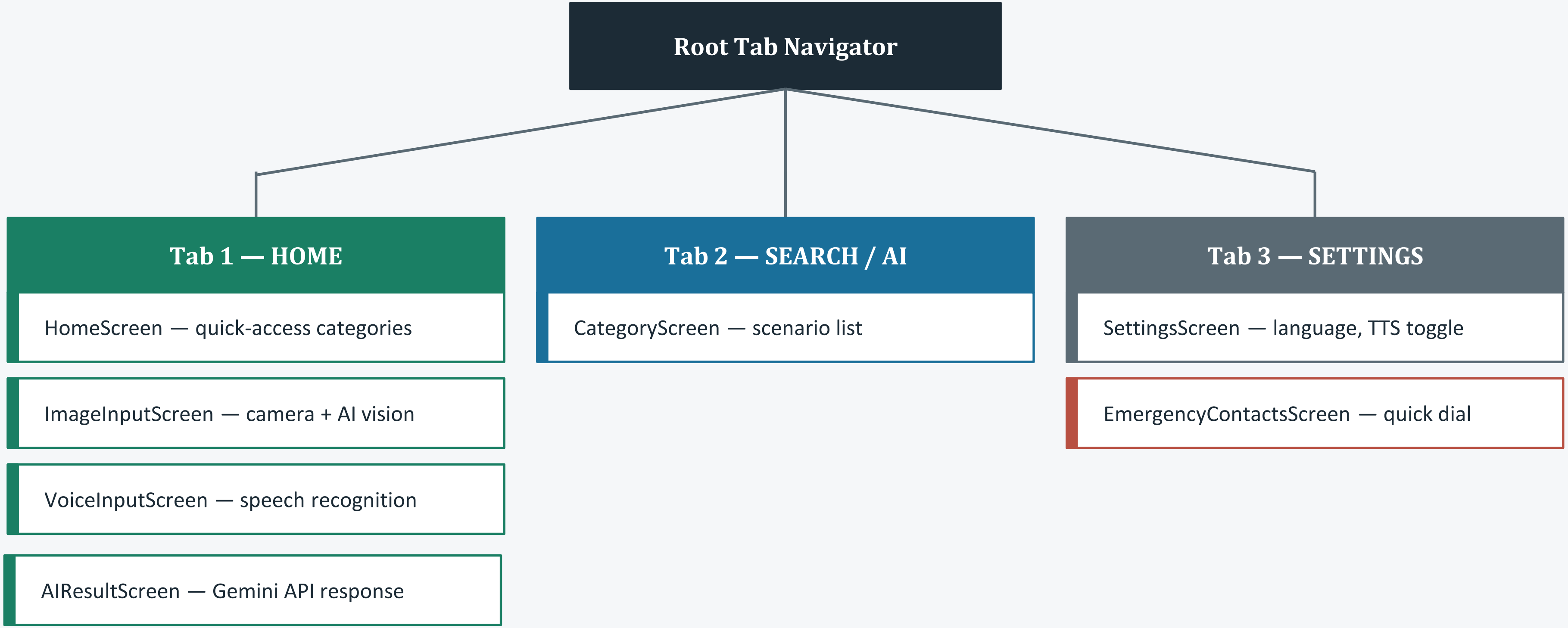
Scenario

Numbered steps in large font with Swahili toggle. Improvised resources accordion. Floating TTS button reads each step aloud and auto-advances — fully hands-free.

Voice Input

Full-screen mic visualiser. Real-time transcription preview. Routes to Claude (online) or offline matcher (offline). Connectivity state always visible.

Fig 5.4 — Navigation Architecture



API Integration — Three Request Modes

All three modes hit the same generateContent endpoint. The parseAIResponse() handles rendering and error fallback.

Text Query

model: gemini-2.5-flash · maxOutputTokens: 4096
systemInstruction: [emergency prompt] · messages: [{role: user, content: userText}]

Speech Query

Identical to text query — the expo-speech-recognition transcript is passed directly as the user message content string.

Image + Text

Content: [{inlineData: {mimeType: image/jpeg, data: base64Str}}, {text: userText}]

Error handling: 429 / 503 / timeout → silently fall back to offline scenario — user always gets actionable guidance.

Chapter 6 - Implementation and Testing Development Environment, System Build, and Quality Evaluation

Development Environment

Hardware and Test Devices

•Primary: Infinix Note 30 Pro (Android 13) — physical device, all sprint testing

•Emulator: Pixel 6 API 33 — Android Studio AVD, baseline regression tests

•Emulator: iPhone 15 Simulator — iOS 17 Xcode, cross-platform smoke tests

Software and Configuration

•React Native 0.81.5 (Expo SDK 54) — cross-platform mobile framework

•Google Gemini API: gemini-2.5-flash (primary) / gemini-1.5-flash (fallback)

•API key loaded from .env via react-native-dotenv — never committed to source control

•SQLite local database — seeded on first boot, initialises within 2 seconds

Implementation Highlights

Offline Database — 50 bilingual emergency scenarios seeded on first boot

•SQLite with 5 tables: categories, scenarios, steps, imp_resources, user_settings

AI Triage Engine — claudeService.js (legacy filename; uses Google Gemini API)

•3-attempt exponential backoff (2 s base delay) before offline fallback on HTTP 429/503

•parseAIResponse() strips markdown and maps response sections to structured UI cards

Swahili TTS — Speech.getAvailableVoicesAsync() selects best sw-KE or sw-TZ voice

•Ordinals: *Hatua ya kwanza / pili / tatu* (not literal digit pronunciation)

•Speaking rate: 0.82 — tuned for clarity under emergency stress conditions

Functional Results

ID	Requirement	Result
FR01–02	Offline scenario guides and improvised resource mappings	PASS
FR03–04	Keyword routing and offline state UI updates	PASS
FR05–06	Online AI text triage and structured response parsing	PASS
FR07–09	Speech recognition and hands-free TTS step narration	PASS
FR10–11	Camera capture and Gemini Vision image analysis	PASS
FR12	One-tap emergency dial with confirmation overlay	PASS

Navigation Audit — All critical paths confirmed within 3 taps from the Home screen

Non-Functional and Usability Results

Performance — PASS

- Offline scenario load: **0.12 s** average (target: ≤ 1.0 s) ✅
- Gemini API response: **3.8 s** average (target: ≤ 5.0 s) ✅

Reliability and Accuracy — PASS / PARTIAL

- Zero crashes across all 50 offline stress runs ✅
- 96% Correct or Partially Correct across 50 AI test descriptions ⚠️ (2 ambiguous multi-symptom edge cases; offline library provided fallback in both)

Usability Study — n=15 representative participants, SUS think-aloud protocol

- Emergency dial task completion: **100%**
- Scenario location within 3 taps: **86.6%** (13 of 15 without assistance)

- **SUS Score: 76.5** — exceeds target of 68, achieves the *Good* usability band

Chapter 7 - Conclusions and Recommendations Objective Achievement, Lessons Learnt, and Future Work

Achievement of Objectives

Objective 1 — Investigate Knowledge Gaps: ACHIEVED

•Structured interviews (n=10) shaped the 50-scenario library and improvised resource engine

Objective 2 — Panic-Proof Interface: ACHIEVED

•Single-purpose Home screen, all critical actions within 3 taps, SUS score 76.5 (Good)

Objective 3 — AI Triage Module: ACHIEVED

•Gemini API integrated with text, voice, and image modalities; 96% accuracy achieved

Objective 4 — Improvised Resource Engine: ACHIEVED

•imp_resources SQLite table maps household items to standard medical supplies per scenario

Objective 5 — Usability Evaluation: ACHIEVED

•15 functional tests (all PASS) and 15-user usability study (SUS 76.5)

Lessons Learnt

Prompt engineering is an iterative discipline

•Reliable structured responses required many refinement cycles; strict formatting rules (no markdown, fixed section labels) were essential to consistent response parsing.

Offline-first must be a founding constraint, not an afterthought

•Building the SQLite tier first prevented major architectural rework in later sprints. Every feature was designed with an offline fallback from day one.

Physical device testing is irreplaceable

•Tab bar inset clipping on the Infinix Note 30 Pro and TTS voice fallback issues were invisible on emulators and would have shipped as bugs without physical device testing.

API key management requires discipline from day one

•Environment variable isolation via .env and react-native-dotenv prevents accidental source control leaks and makes production key rotation straightforward.

Recommendations for Future Work

Play Store and App Store Publication

•Move from APK sideloading (Android) and TestFlight (iOS) to full store listings for wider reach, auto-updates, and compliance with platform security requirements.

API Key Proxy Backend

•Route all Gemini API calls through a Cloud Function or Firebase Function to protect the API key from binary extraction from the production APK.

Offline AI via On-Device Models

•Explore Gemma 3 or Phi-3 Mini integration to provide AI-quality guidance without internet connectivity, eliminating the dependency on the Gemini API for critical use cases.

Clinical Scenario Validation

•Partner with Kenya Red Cross and clinical professionals to review, validate, and expand the scenario library beyond the initial 50, with a community contribution workflow.

Conclusion

AidField successfully closes a critical gap in emergency response technology

•Delivers reliable offline guidance for 50 evidence-based scenarios with zero connectivity

•Enhances guidance with real-time AI triage, voice, and visual input when online

The dual-tier architecture ensures no user is ever left without actionable guidance

•*Online tier:* Gemini API multimodal triage (text, voice, image), 96% accuracy

•*Offline tier:* SQLite library with bilingual steps and improvised resource substitutes

Broader Contribution to Research and Practice

•Advances mHealth design for low-resource environments in alignment with SDG 3 (Good Health and Well-Being)

•Provides a replicable, validated framework for emergency health technology across East Africa

•*"In its production form, AidField has the potential to save lives."*